

The University of Nottingham

SCHOOL OF COMPUTER SCIENCE

A LEVEL 1 MODULE, SPRING SEMESTER 2020-2021

Mathematics for Computer Scientists 2

Time allowed: X Hours and Y Minutes

Candidates may complete the front cover of their answer book and sign their desk card but must NOT write anything else until the start of the examination period is announced

«Answer Rubric»

«Calculator Rubric»

Dictionaries are not allowed with one exception. Those whose first language is not English may use a standard translation dictionary to translate between that language and English provided that neither language is the subject of this examination. Subject specific translation dictionaries are not permitted.

No electronic devices capable of storing and retrieving text, including electronic dictionaries, may be used.

DO NOT turn examination paper over until instructed to do so

ADDITIONAL MATERIAL: «List Additional/Supplementary Materials MUST be specified in this section of the paper Handouts, MCQ, number of answer books x , graph paper, statistical tables (Neaves, SI, E&E), statutes, Books allowed, None».

INFORMATION FOR INVIGILATORS: «to include any announcements or to indicate if the paper can be taken away at the end of the examination / None».

SECTION A

QUESTION 1**1. Set Theory**

Calculate $A \times B$ where

$$A = \{a, b, c\}$$

and

$$B = \{x \in \mathbb{Z} \mid x^2 - 2x - 8 = 0\}.$$

[< < > > Marks]

Solution

$$B = \{-2, 4\}$$

$$A \times B = \{(a, -2), (a, 4), (b, -2), (b, 4), (c, -2), (c, 4)\}$$

QUESTION 2**2. Matrices**

Let us consider the following matrix

$$\begin{pmatrix} 1 & -2 & 0 \\ 2 & 2 & 6 \\ 0 & 1 & 1 \end{pmatrix}$$

- Determine the rank of the matrix

[< < > > Marks]

- Invert the matrix if possible. If it is impossible, justify your answer.

[< < > > Marks]

End of Question 2: Total < < > > marks

Solution

If the matrix is represented as $\mathbf{A} = (\mathbf{a}^1, \mathbf{a}^2, \mathbf{a}^3)$, then $\mathbf{a}^3 = 2\mathbf{a}^1 + \mathbf{a}^2$.

- Thus the rank of the matrix is 2
- The matrix is singular and cannot be inverted

QUESTION 3**3. Gaussian elimination**

Let us consider the following system of linear equations

$$\begin{cases} x - y + z = 1 \\ x + y = 4 \\ 2x + 2y + 2z = 9 \end{cases}.$$

- Apply, if possible, the Gaussian elimination to find the equivalent triangular matrix/system

[< < > > Marks]

- Use the Gaussian elimination for performing the LU factorization

[< < > > Marks]

- With the detected triangular system find the solution of the system

[< < > > Marks]

End of Question 3: Total < < > > marks**Solution**

- We can see the $\rho_{\mathbf{A}} = \rho_{\mathbf{A}^c} = 3$. Hence we can apply Gaussian elimination. For the complete matrix

$$\mathbf{A}^c = (\mathbf{A}|\mathbf{b}) = \left(\begin{array}{ccc|c} 1 & -1 & 1 & 1 \\ 1 & 1 & 0 & 4 \\ 2 & 2 & 2 & 9 \end{array} \right).$$

Let us at first apply the following row operations:

$$\mathbf{r}^2 = \mathbf{r}^2 - \mathbf{r}^1$$

$$\mathbf{r}^3 = \mathbf{r}^3 - 2\mathbf{r}^1.$$

We obtain

$$\mathbf{A}^c = (\mathbf{A}|\mathbf{b}) = \left(\begin{array}{ccc|c} 1 & -1 & 1 & 1 \\ 0 & 2 & -1 & 3 \\ 0 & 4 & 0 & 7 \end{array} \right).$$

Now, let us apply the following row transformation

$$\mathbf{r}^3 = \mathbf{r}^3 - 2\mathbf{r}^2.$$

Thus, we obtain the triangular matrix

$$\mathbf{A}^c = (\mathbf{A}|\mathbf{b}) = \left(\begin{array}{ccc|c} 1 & -1 & 1 & 1 \\ 0 & 2 & -1 & 3 \\ 0 & 0 & 2 & 1 \end{array} \right).$$

- From the calculations above we can calculate

$$\mathbf{L} = \left(\begin{array}{ccc} 1 & 0 & 0 \\ 1 & 1 & 0 \\ 2 & 2 & 1 \end{array} \right)$$

and

$$\mathbf{U} = \left(\begin{array}{ccc} 1 & -1 & 1 \\ 0 & 2 & -1 \\ 0 & 0 & 2 \end{array} \right)$$

- The triangular system is

$$\begin{cases} x - y + z = 1 \\ 2y - z = 3 \\ 2z = 1 \end{cases}.$$

We can find $z = \frac{1}{2}$, $2y - \frac{1}{2} = 3$ that is $y = \frac{7}{4}$. By substituting $x - \frac{7}{4} + \frac{1}{2} = 1$ that $x = \frac{9}{4}$.

QUESTION 4

4. Vector Spaces

Let us consider the vector space $(U, +, \cdot)$ where

$$U = \left\{ (x, y, z) \in \mathbb{R}^3 \mid \begin{cases} x + 2y - z = 0 \\ 2x + 4y - 2z = 0 \\ 6x + 12y - 6z = 0 \end{cases} \right\}.$$

- Determine a basis of the vector space

[< < > > Marks]

- Determine the dimension of the vector space

[< < > > Marks]

- Let $(V, +, \cdot)$ be a vector space whose basis $B_V = \{(0, 0, 0)\}$. Verify the Grassmann's formula for these two vector spaces.

[< < > > Marks]

- Provide a geometric interpretation of what Grassmann's formula states for U and V

[< < > > Marks]

End of Question 4: Total < < > > marks

Solution

- We can see that the equations are all proportional. Hence the rank is 1. Thus, if we pose $y = \alpha$ and $z = \beta$ we obtain $x = \beta - 2\alpha$. This means that from the general solution we can find the basis of U

$$(\beta - 2\alpha, \alpha, \beta) = \alpha(-2, 1, 0) + \beta(1, 0, 1)$$

that is

$$B_U = \{(-2, 1, 0), (1, 0, 1)\}$$

- Since the basis has two vector its dimension is 2
- We can see that since $B_V = \{(0, 0, 0)\}$ then $U \cap V = \{(0, 0, 0)\}$. This is because vector spaces always contain at least the null vector. Also, $U + V = U$. We can thus verify

$$\dim(U + V) = \dim(U) + \dim(V) - \dim(U \cap V) = 2 + 0 - 0 = 2.$$

- U is a plane of the space while V is only the null the vector. The intersection of these two sets is only the null vector. The sum set of these vectors is the plane identified by U since the contribution of V (null vector) does not affect the geometry of $U + V$.

SECTION B

QUESTION 1**1. Matrices**

Let $\alpha, \beta \in \mathbb{R}$. Let us consider the following matrix

$$\mathbf{A} = \begin{pmatrix} a_{11} & a_{12} & \alpha a_{11} + \beta a_{12} \\ a_{21} & a_{22} & \alpha a_{21} + \beta a_{22} \\ a_{31} & a_{32} & \alpha a_{31} + \beta a_{32} \end{pmatrix}$$

with $a_{ij} \in \mathbb{R}$ and $a_{ij} \neq 0, \forall i, j$. Let us now consider a matrix $\mathbf{B}$ such that

$$\mathbf{AB} = \mathbf{O}$$

with $\mathbf{O}$ null matrix.

- Determine the possible rank values of $\mathbf{B}$. Justify your answer and use the relevant theory.

[< < > > Marks]

- Can it ever happen that $\mathbf{B}$ is non-singular? Justify your answer.

[< < > > Marks]

- Can it ever happen that $\mathbf{B}$ is the null matrix? Justify your answer.

[< < > > Marks]

End of Question 1: Total < < > > marks

Solution

- We know that $\mathbf{A}$ is not the null matrix and is singular. The matrix $\mathbf{A}$ can be of rank 1 or 2. When know that $\rho_{\mathbf{AB}} = 0$ From Sylvester's nullity theorem we have two options. If $\rho_{\mathbf{A}} = 1$, $0 \geq 1 + \rho_{\mathbf{B}} - 3$. This means that $\rho_{\mathbf{B}} \leq 2$, that is the possible ranks are 0, 1, 2. If $\rho_{\mathbf{A}} = 2$, $0 \geq 2 + \rho_{\mathbf{B}} - 3$. This means that $\rho_{\mathbf{B}} \leq 1$, that is the possible ranks are 0, 1.
- From Sylvester's nullity theorem we determined that $\rho_{\mathbf{B}} < 3$ in all scenarios. Thus, $\mathbf{B}$ is surely singular.
- From Sylvester's nullity theorem we determined that $\rho_{\mathbf{B}}$ could be 0. Thus, it could be the null matrix.

QUESTION 2**2. Vector Spaces/Linear Mappings**

Let $(U, +, \cdot)$ be a vector subspace of $(\mathbb{R}^n, +, \cdot)$ with $\dim(U) < n$. Let $f : U \rightarrow \mathbb{R}^n$ be a linear mapping.

- Assess whether or not the mapping can be injective. Justify your answer by using the relevant theorems.

[< < > > Marks]

- Assess whether or not the mapping can be surjective. Justify your answer by using the relevant theorems.

[< < > > Marks]

- Assess whether or not the mapping can be invertible. Justify your answer by using the relevant theorems.

[< < > > Marks]

End of Question 2: Total < < > > marks

Solution

- The mapping could be injective as long as the rank of the matrix associated with it is equal to its number of columns. In that case the kernel would be only the null vector and then the mapping would be injective.
- Since $\dim(U) < n = \dim(\mathbb{R}^n)$ the dimensions of domain and codomain are different. Hence for the corollary 10.4 of the Rank-Nullity Theorem, the mapping cannot be surjective.
- Only bijective mappings are invertible for Proposition 10.6. Since the mapping is not surjective will also be not invertible.